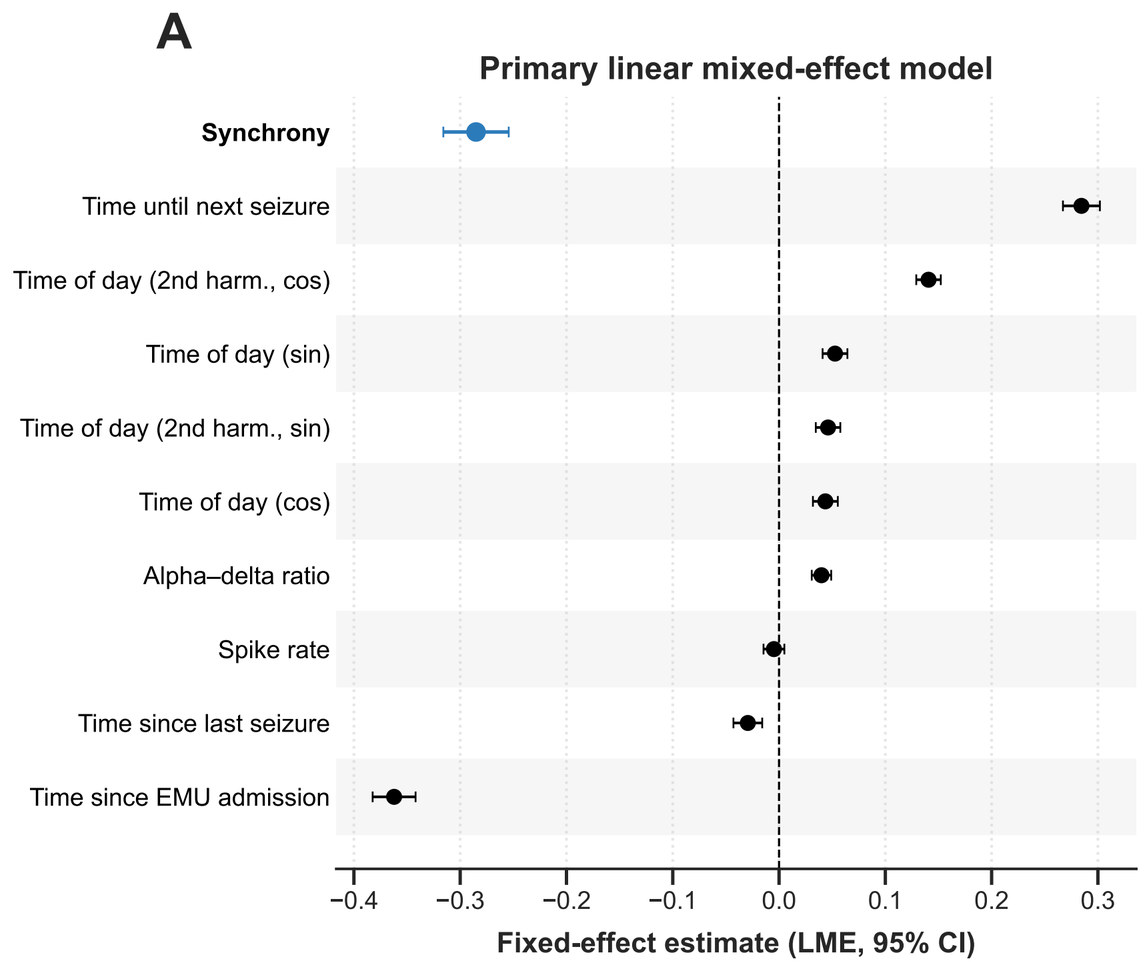
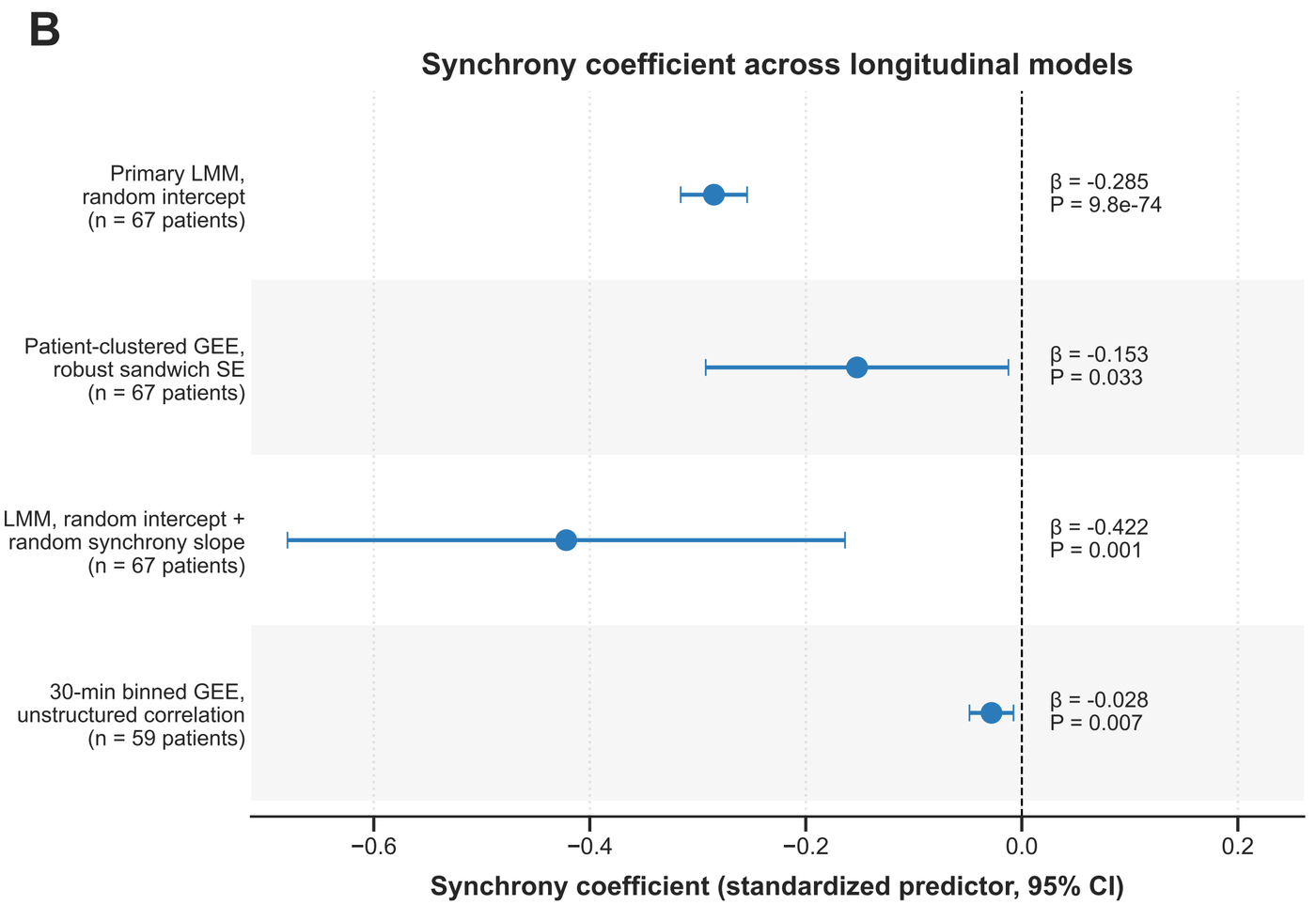
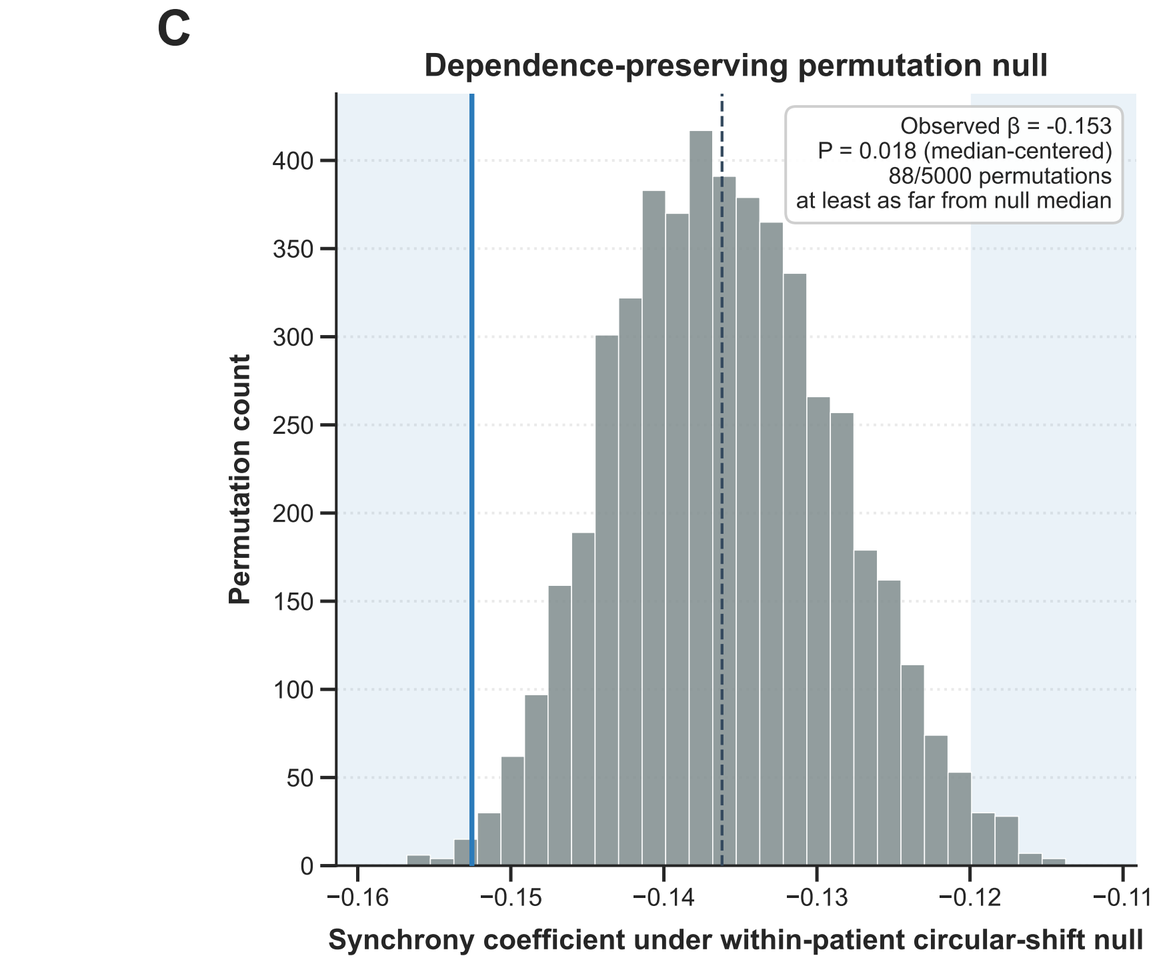
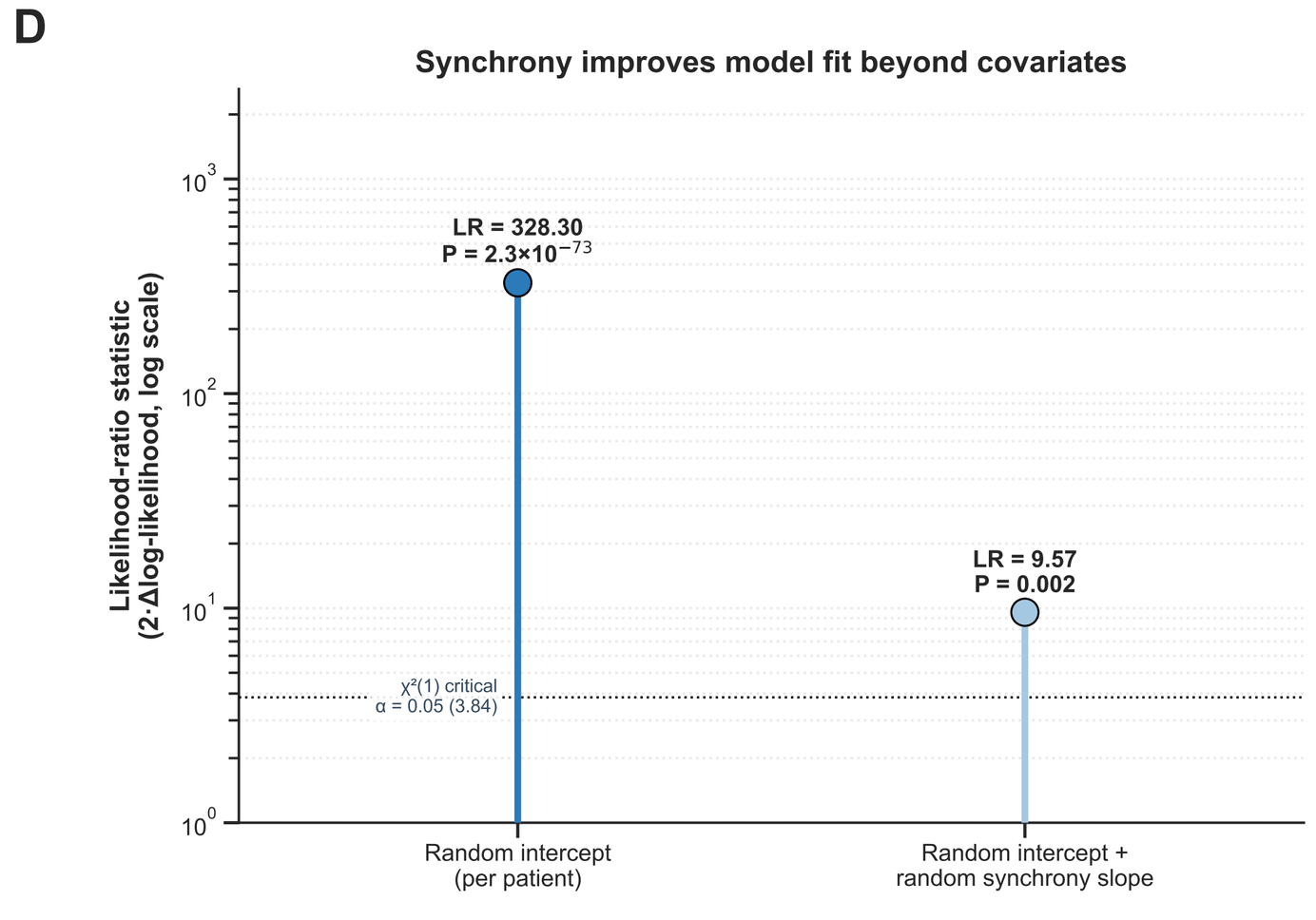

A
Primary linear mixed-effect model
Synchrony
Time until next seizure
Time of day (2nd harm., cos)
Time of day (sin)
Time of day (2nd harm., sin)
Time of day (cos)
Alpha–delta ratio
Spike rate
Time since last seizure
Time since EMU admission
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Fixed-effect estimate (LME, 95% CI)


B
Synchrony coefficient across longitudinal models
Primary LMM,
random intercept
(n = 67 patients)
β = -0.285
P = 9.8e-74
Patient-clustered GEE,
robust sandwich SE
(n = 67 patients)
β = -0.153
P = 0.033
LMM, random intercept +
random synchrony slope
(n = 67 patients)
β = -0.422
P = 0.001
30-min binned GEE,
unstructured correlation
(n = 59 patients)
β = -0.028
P = 0.007
−0.6
−0.4
−0.2
0.0
0.2
Synchrony coefficient (standardized predictor, 95% CI)


C
Dependence-preserving permutation null
Observed β = -0.153
P = 0.018 (median-centered)
88/5000 permutations
at least as far from null median
Permutation count
0
50
100
150
200
250
300
350
400
−0.16
−0.15
−0.14
−0.13
−0.12
−0.11
Synchrony coefficient under within-patient circular-shift null


D
Synchrony improves model fit beyond covariates
Likelihood-ratio statistic
(2·Δlog-likelihood, log scale)
10^3
10^2
10^1
10^0
LR = 328.30
P = 2.3×10^−73
LR = 9.57
P = 0.002
χ²(1) critical
α = 0.05 (3.84)
Random intercept
(per patient)
Random intercept +
random synchrony slope